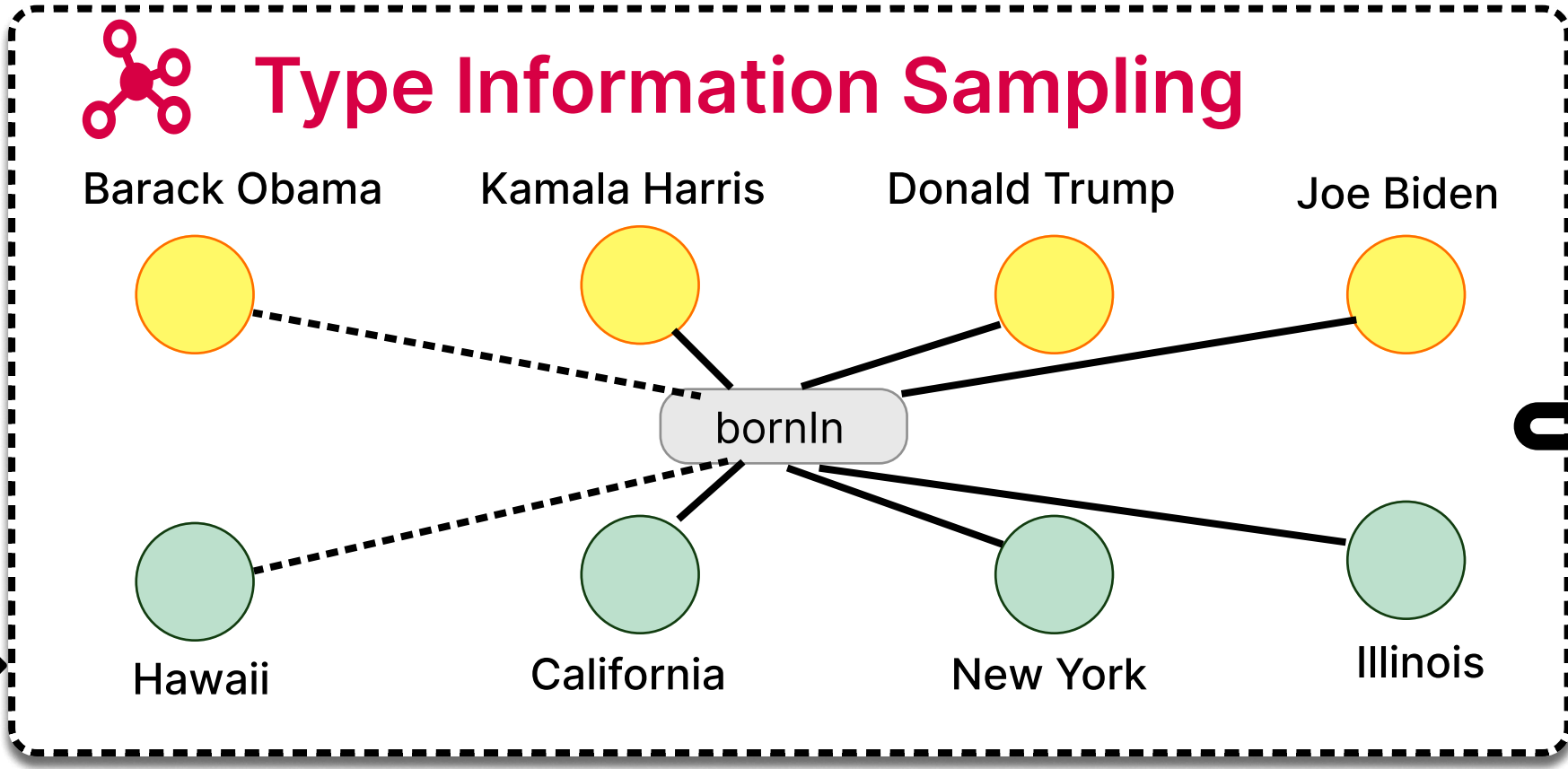
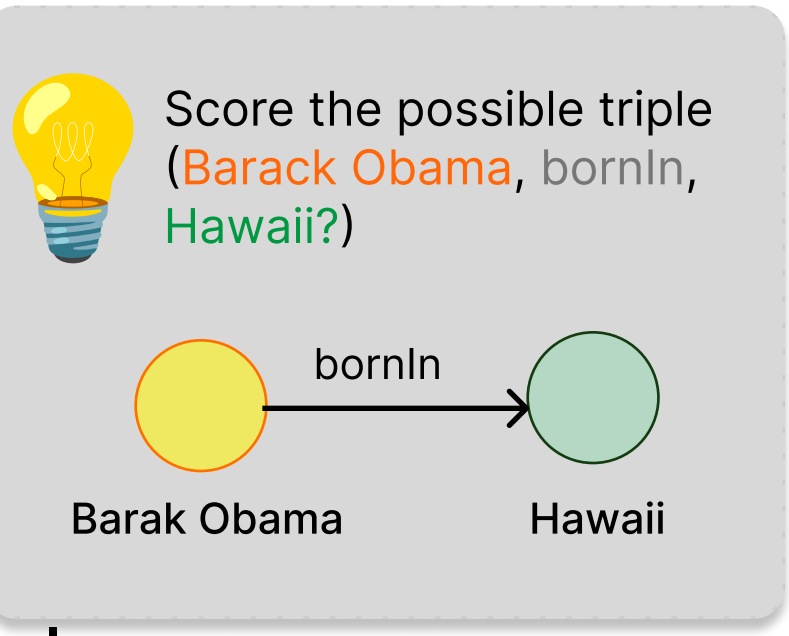




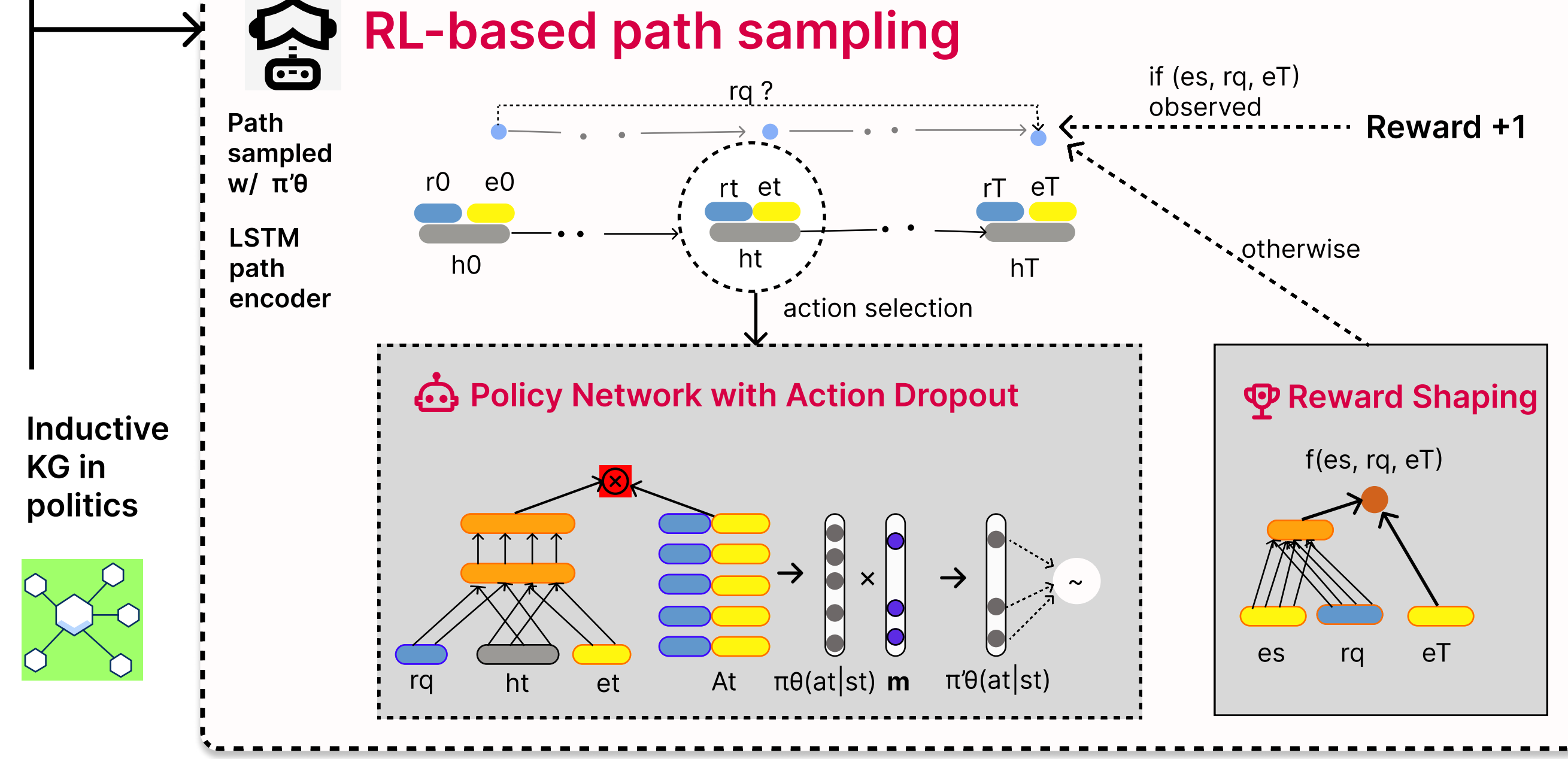
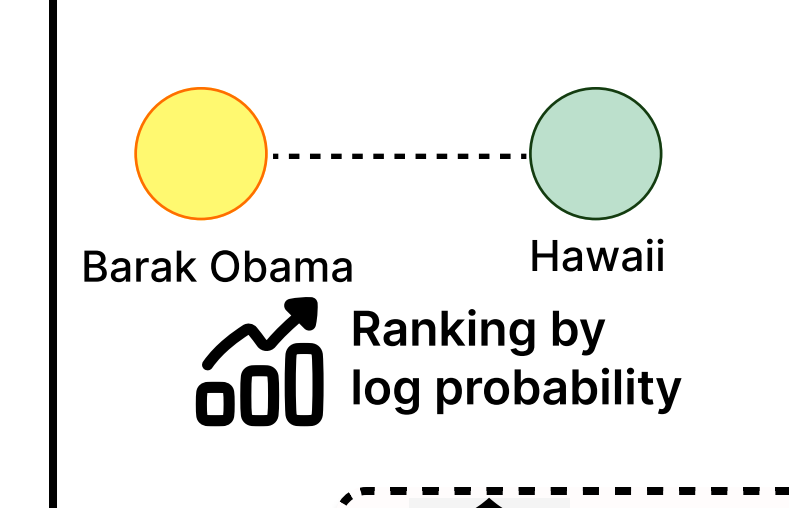
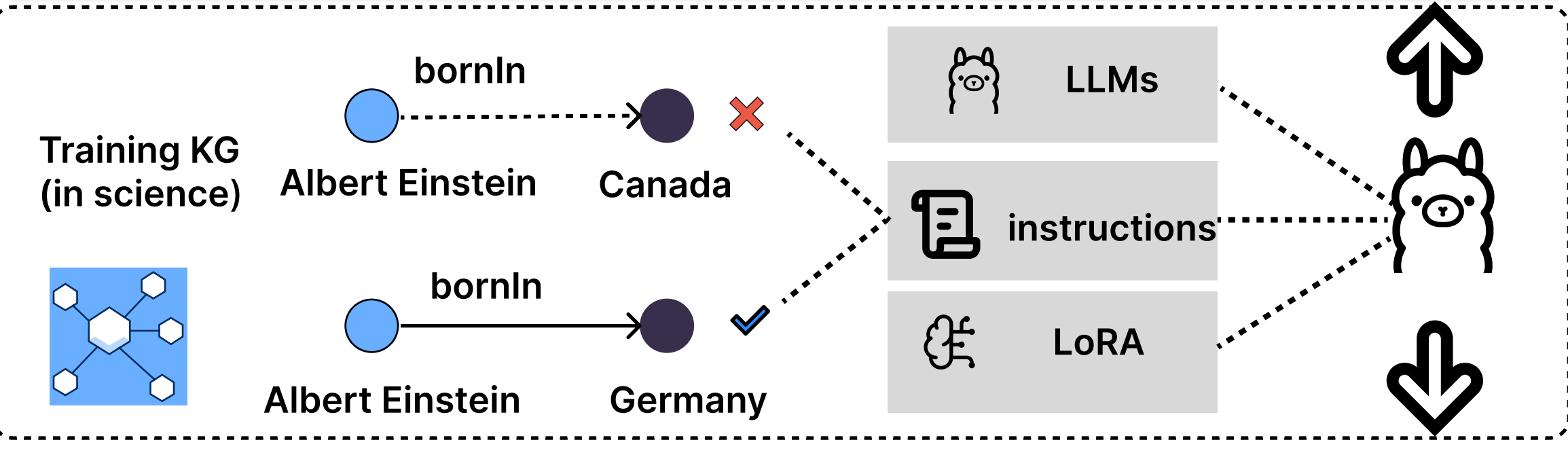
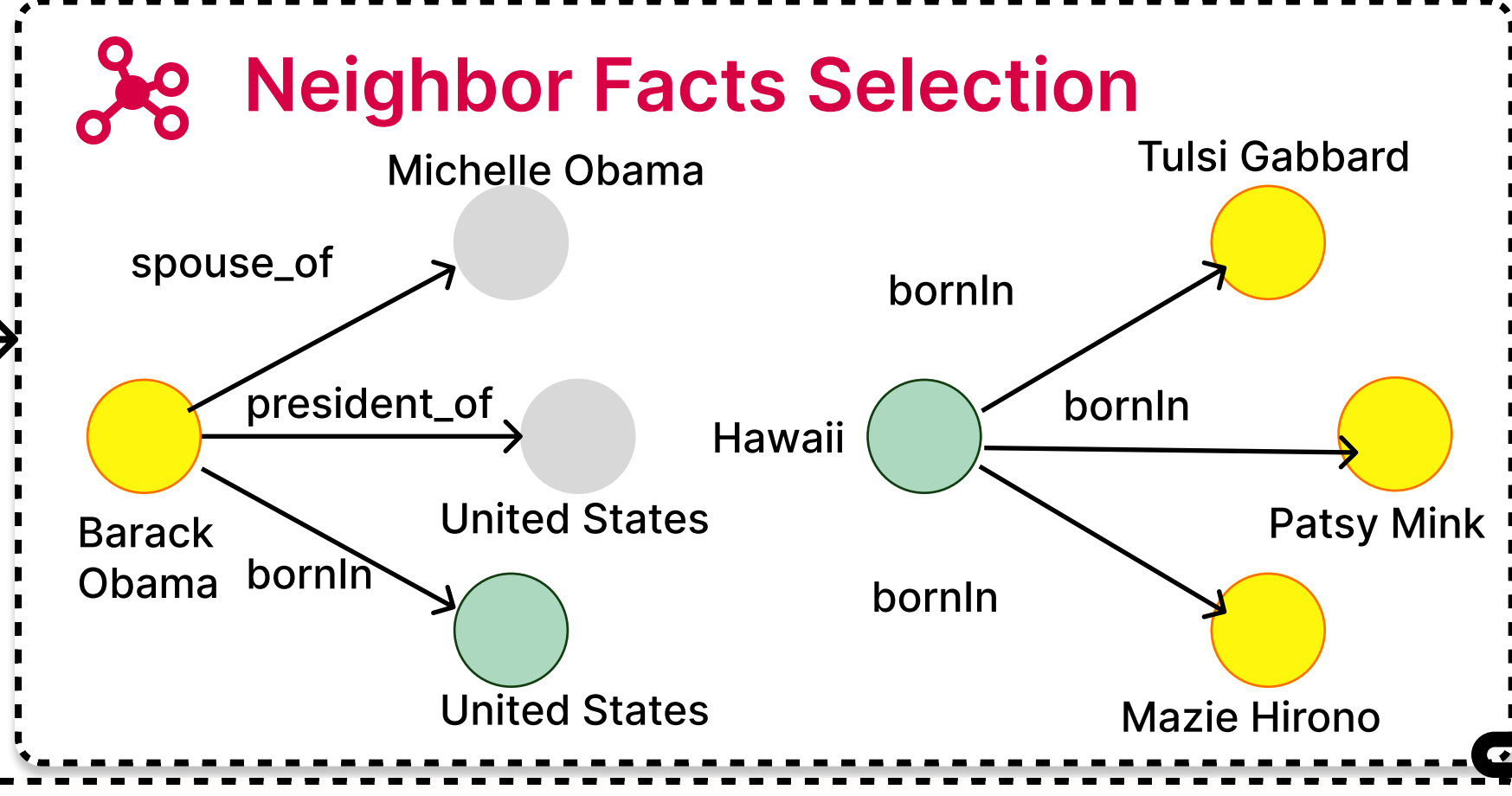
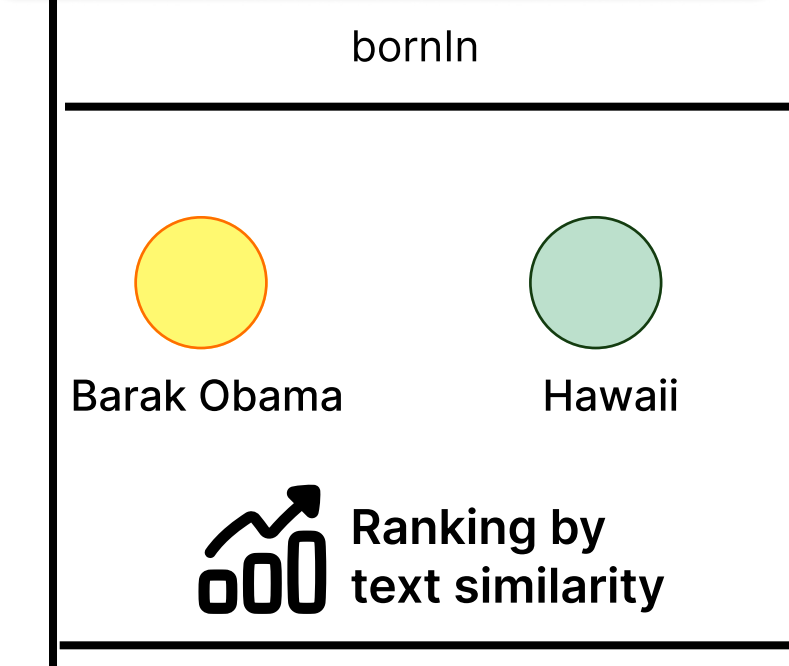
Type-aware reasoning (TAR)

Please determine whether the entities in the input triples are consistent in **entity type** with a set of known triples in the knowledge graph provided.

A set of known triples are:
(**'Kamala Harris'** was born in **'California'**)
(**'Joe Biden'** was born in **'Pennsylvania'**)
(**'Donald Trump'** was born in **'New York'**)

The triple to be determined is:
(**'Barack Obama'** was born in **'Hawaii'**)

Please return 'Y' if the input triple is consistent in **entity type**, otherwise return 'N'. Do not say anything else except your determination.



Subgraph Reasoning (SR)

Please determine whether the relation in the input can be reliably inferred between the head and tail entities, based on a set of **neighbor triples and relation paths** from the knowledge graph.

A set of **neighbor triples** from the knowledge graph are:
(**'Barack Obama'**, spouseOf, 'Michelle Obama')
(**'Barack Obama'**, presidentOf, 'United States')
(**'Barack Obama'**, bornIn, 'United States')
(**'Tulsi Gabbard'**, bornIn, **'Hawaii'**)

A set of **relation paths** from the knowledge graph are:
(**'Barack Obama'**, bornIn, 'Honolulu') $\rightarrow$ ('Honolulu', locatedIn, **'Hawaii'**)
(**'Barack Obama'**, educatedAt, 'Punahou School') $\rightarrow$ ('Punahou School', locatedIn, 'Honolulu') $\rightarrow$ ('Honolulu', locatedIn, **'Hawaii'**)
(**'Barack Obama'**, livedIn, 'Honolulu') $\rightarrow$ ('Honolulu', locatedIn, **'Hawaii'**)

The relation to be inferred is:
(**'Barack Obama'**, bornIn, **'Hawaii'**)

Please return 'Y' if there is sufficient evidence from the knowledge graph to infer the relation, otherwise return 'N'.

